

1. Administrative & Study Identification

Overall Study Title: A Study to Evaluate Efficacy, Safety & Pharmacokinetics of the Port Delivery System (PDS) With Ranibizumab in Participants With Diabetic Macular Edema (DME) Compared With Intravitreal Ranibizumab; A Substudy to Evaluate the Safety of Re-implanting the PDS With Ranibizumab in Participants With DME **Official Title:** A Phase III, Multicenter, Randomized, Visual Assessor-Masked, Active-comparator Study of the Efficacy, Safety, and Pharmacokinetics of the Port Delivery System With Ranibizumab in Patients With Diabetic Macular Edema (Pagoda) **Short Title/Acronym:** Pagoda **Protocol Number:** GR40550 **NCT Number:** NCT04108156 **Sponsor/Company:** Roche (Hoffmann-La Roche) **Trial Status:** Recruiting **Investigational Product:** Port Delivery System (PDS) Implant Pre-Filled with 100 mg/mL ranibizumab (Trade names: Lucentis, Susvimo, Cimerli, Byooviz) **Phase of Development:** Phase III **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead (Global)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy, safety, and pharmacokinetics (PK) of continuous ranibizumab (100 mg/mL) delivery via the Port Delivery System (PDS) with refill exchanges every 24 weeks (Q24W) compared to intravitreal (IVT) ranibizumab (0.5 mg) injections every 4 weeks (Q4W) in participants with DME. **Substudy Objective:** To evaluate the safety of re-implanting the updated PDS with ranibizumab and the refill-exchange procedures following re-implantation in up to 100 participants with DME who were previously enrolled in the main study (GR40550). **Secondary Objectives:** To evaluate changes in Best-Corrected Visual Acuity (BCVA) across different statistical strategies, improvements in the Early Treatment Diabetic Retinopathy Study (ETDRS) – Diabetic Retinopathy Severity Score (DRSS), and overall safety parameters.

2.2 Background & Rationale Frequent visits and intravitreal anti-vascular endothelial growth factor (VEGF) injections are typically required to manage diabetic macular edema (DME), which creates a significant treatment burden for patients and healthcare systems. The Port Delivery System (PDS) with ranibizumab is designed as a continuous anti-VEGF therapy that aims to reduce visit and treatment burdens by maintaining therapeutic drug concentration levels with only two refills per year, without sacrificing vision outcomes.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator **Blinding:** Visual Assessor-Masked **Randomization:** Randomized (Parallel Assignment)

3.2 Planned Enrollment & Duration Target N: 634 participants
Number of Sites: Approximately 87–102 sites **Estimated Study Start:** 30-Sep-2019 **Estimated Primary Completion:** 16-Nov-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria (Main Study) 1. Age ≥ 18 years at the time of signing informed consent. 2. Documented diagnosis of diabetes mellitus (Type 1 or Type 2) with a glycated hemoglobin (HbA1c) level of $\leq 10\%$ within 2 months prior to or at screening. 3. Study Eye: Macular thickening secondary to DME involving the center of the fovea with Central Subfield Thickness (CST) ≥ 325 μm on SD-OCT at screening, and a BCVA score of 78 to 25 letters (20/32 to 20/320 approximate Snellen equivalent).

4.2 Exclusion Criteria (Main Study) 1. High-risk Proliferative Diabetic Retinopathy (PDR). 2. Active intraocular inflammation (grade trace or above) or suspected/active ocular or periocular infection of either eye. 3. Uncontrolled ocular hypertension or glaucoma that may require glaucoma-filtering surgery during the study. 4. Cerebrovascular accident or myocardial infarction within 6 months prior to randomization. 5. Atrial fibrillation diagnosis or worsening within 6 months prior to randomization. 6. Uncontrolled blood pressure.

4.3 Substudy Eligibility Criteria Substudy Inclusion Criteria: 1. Having experienced a septum dislodgement in the original implant while in the main study or after exiting the main study. 2. Sufficiently clear ocular media and adequate pupillary dilation to allow for analysis and grading by a central reading center.

Substudy Exclusion Criteria (Cohort 1 Only / Ocular): 1. Recent history (last 3 months) of other diseases/conditions contraindicating ranibizumab or PDS placement, active cancer within the last 12 months (with exceptions for appropriately treated in situ cervix, non-melanoma skin, or prostate cancer), or current systemic treatment for confirmed active systemic infection. 2. Participation in another investigational trial within 6 months or use of antimitotic/antimetabolite therapy within 30 days. 3. Intraocular surgery (including cataract surgery) within 1 month preceding enrollment. 4. History of rhegmatogenous retinal tears/breaks within 3 months, concurrent ocular conditions requiring surgical intervention, or concurrent structural conditions in the supero-temporal quadrant that may affect refill-exchange. 5. Use of medicated intraocular implants (other than PDS) at any time prior to enrollment. 6. Suspected/

active ocular or periocular infection, active blepharitis, or any history of uveitis (in either eye).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - **PDS Arm:** Surgical insertion of PDS implant (pre-filled with ranibizumab 100 mg/mL) after an initial loading phase, with refill-exchange procedures performed at a fixed interval of every 24 weeks (Q24W). - **Active Comparator Arm:** Intravitreal (IVT) ranibizumab 0.5 mg injections every 4 weeks (Q4W). **Route of Administration:** Ocular Implant (PDS) / Intravitreal Injection **Duration of Treatment:** 64 weeks for the primary core phase; substudy participants will be followed for a maximum of 72 weeks post-re-implantation.

5.2 Concomitant & Prohibited Therapy Permitted: Standard systemic treatments for diabetes, hypertension, and routine medical care. **Prohibited:** Concurrent use of medicated intraocular implants (other than the PDS implant) or any concurrent ocular condition/treatment that would require surgical intervention during the study to prevent or treat visual loss; antimitotic or antimetabolite therapy (within 30 days of substudy).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical/Ocular History, OCT, BCVA, HbA1c Lab
Baseline	Day 1	Randomization, First IVT Injection or PDS loading injection
Interim	Weeks 4 to 60	Safety Assessments, BCVA tracking, PDS surgical insertion, Refill-exchanges every 24 Weeks (Q24W)
End of Study	Week 64 / Extended Follow-Up	Final BCVA and OCT Assessments, Exit Interview, Safety follow-up (up to 72 weeks post-re-implantation for substudy)

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. **Main Study Primary Endpoint:** Change in BCVA Score from baseline averaged over weeks 60 and 64 as measured using the ETDRS chart in the Efficacy Population using a Treatment Policy Strategy for all intercurrent events. 2. **Substudy Primary Endpoint 1:** Number of participants with ocular and systemic (non-ocular) Adverse Events (AEs) and severity of ocular and systemic AEs. 3. **Substudy Primary Endpoint 2:** Number of participants with Adverse Events of Special Interests (AESIs) and severity of AESIs.

7.2 Secondary Outcomes 1. Change in BCVA Score from baseline averaged over weeks 60 and 64 as measured with use of the ETDRS Chart in the Modified Intent-to-Treat (mITT) Population using a

Treatment Policy Strategy for all intercurrent events. 2. Change in BCVA Score from baseline averaged over weeks 60 and 64 as measured with use of the ETDRS Chart in the mITT Population using a Hypothetical Strategy for all intercurrent events. 3. Percentage of participants with a ≥ 2 -step improvement from baseline on the ETDRS-DRSS at Week 64 in the Efficacy Population.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring for general AEs and ocular Adverse Events of Special Interest (AESIs), including endophthalmitis, retinal detachment, vitreous hemorrhage, and implant dislocation/septum dislodgement. **Serious Adverse Events (SAEs):** Assessed and documented continuously; reported immediately to the sponsor and Independent Data Monitoring Committee (IDMC). **Data Monitoring Committee (DMC):** An independent data monitoring committee routinely reviewed safety data throughout the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Efficacy population and Modified Intent-to-Treat (mITT) population for primary and secondary efficacy outcomes; Safety Set for all adverse events. **Sample Size Justification:** Target enrollment of 634 participants, sized to demonstrate non-inferiority of the PDS Q24W against monthly ranibizumab using a pre-specified non-inferiority margin. **Handling of Missing Data:** Evaluated utilizing Treatment Policy Strategies and Hypothetical Strategies to address intercurrent events and missing data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with the Declaration of Helsinki and International Conference on Harmonisation (ICH) E6 guidelines for Good Clinical Practice (GCP). **Monitoring Plan:** Rigorous onsite and masked visual assessor auditing. Institutional Review Board (IRB) approval was secured for all trial sites. **Audit Trail:** Secure Electronic Data Capture (EDC) systems utilized with regular data clarification checks to ensure masking integrity.

11. Study Identifiers & Governance

Primary ID: {{GR40550}} **Registry Number:** {{NCT04108156}} **Posting ID:** {{1026602851728017749}} **Sponsor/Company:** {{Roche}} **Overall Study Title:** {{A Study to Evaluate Efficacy, Safety & Pharmacokinetics of the Port Delivery System (PDS) With

Ranibizumab in Participants With Diabetic Macular Edema (DME) Compared With Intravitreal Ranibizumab; A Substudy to Evaluate the Safety of Re-implanting the PDS With Ranibizumab in Participants With DME}} **Official Regulatory Title:** {{A Phase III, Multicenter, Randomized, Visual Assessor-Masked, Active-comparator Study of the Efficacy, Safety, and Pharmacokinetics of the Port Delivery System With Ranibizumab in Patients With Diabetic Macular Edema (Pagoda)}}

12. Clinical Framework (DME Specific)

Primary Condition: {{Diabetic Macular Edema (DME) (MeSH: D000085890, SNOMED: 312912001)}} **Diagnosis Threshold:** {{CST \$\\ge\$ 325\$ μm on SD-OCT with BCVA 78-25 letters}} **Medical Methodology:** {{Continuous Anti-VEGF Therapy}} **Optimization Target:** {{Maintenance of BCVA and reduction in treatment burden}}

13. Study Procedures & Methodology

Management Pathway: {{Surgical PDS insertion followed by Q24W Refill-Exchange; Re-implantation available for eligible substudy participants}} **Education Protocol (HCP):** {{Surgical placement, refill, and exchange protocol training}} **Education Protocol (Patient):** {{Post-operative care and symptom monitoring for ocular AESIs}} **Quality Audit Mechanism:** {{Visual Assessor Masking, Central Reading Center grading, and independent IDMC reviews}}

14. Observation & Follow-Up Schedule

Total Duration: {{64 weeks for main primary core phase; Up to 72 weeks post-re-implantation for substudy}} **Onsite Visit Cadence:** {{Q4W intervals for standard checks; Q24W for PDS refills}} **Remote Monitoring Frequency:** {{As clinically indicated}} **Checklist Review Interval:** {{Every visit}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	____	[DD-MMM-YYYY]				